

# CS & IT ENGINEERING



## 'C' Programming

### Control Statements

Lecture No.- 03



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# Recap of Previous Lecture



- Nested if
- if-else
- Nested if-else
- if-else-if ladder



# Topics to be Covered



- Switch-Case
- do-while
- while
- for

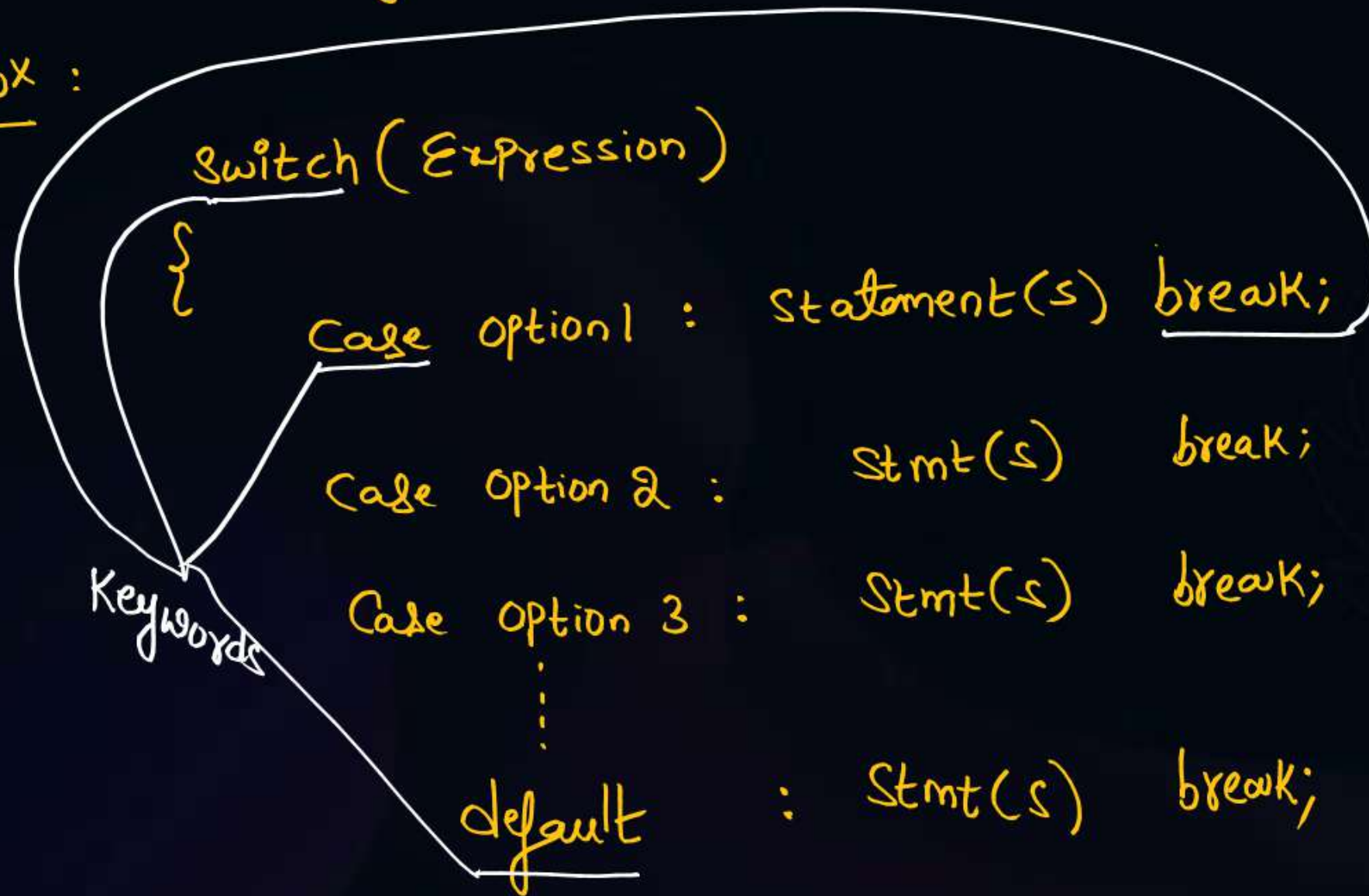




### Switch-Case

- multi-way Selection Statement. [one (or) multiple (or) All]

Syntax :



Ex:

```
int x;
scanf("%d", &x);
switch(x)
{
    case 1 : pf("Hai"); break;
    case 2 : pf("Hello"); break;
    case 3 : pf("How are you?");
              break;
    default : pf("Bye"); break;
}
```

An arrow points from the closing curly brace '}' to the 'break;' statement in the default case.





## Topic : Control Statements - 3



### \*\*\* Important Points about Switch

- ① Expression of switch, must return an integer (or) Integral Expression [int (or) char result  
[ASCII value]]
  - ② default, break are optional
  - ③ default can be any case. [First/middle/last]
  - ④ Case values also must be integral values [int (or) char]
  - ⑤ Cases can be in random order
  - ⑥ Cases can be Non-Contiguous [1, 15, 37, 49, 103]
  - ⑦ Cases can also be Empty.
  - ⑧ Cases must be constants [cannot be variables]
- 9) Duplicate cases are NOT allowed.  
[in either integer (or) char form]



## Topic : Control Statements - 3



### Examples

①

```
int a=1, b=2, c=3, d=4, e=5, x;  
scanf("%d", &x);  
switch(x)  
{  
    case a: Pf("Hai"); break;  
    case b: Pf("Hello"); break;  
    case c: Pf("Hi"); break;  
    case d: Pf("Hey"); break;  
    case e: Pf("How are you"); break;  
    default: Pf("Bye"); break;  
}  
// Error, cases cannot be variables.
```

②

```
float x;  
scanf("%f", &x);  
switch(x)  
{  
    case 1.1: Pf( )  
    case 1.2: Pf( )  
    case 1.5: Pf( )  
    case 3.4: Pf( )  
    case 7.5: Pf( )  
    default: Pf( )  
}
```

// Error, Exp and values must be integer/character.





## Topic : Control Statements - 3



3)

```
int x;
```

```
scanf("%d", &x);
```

```
switch(x)
```

```
{  
  case 31: Pf("Blue");
```

```
  case 27:
```

```
    default : Pf("Black");
```

```
    case 3 : Pf("Yellow"); break;
```

```
    case 11 : Pf("Orange");
```

```
    case 67 : Pf("white"); break;
```

```
    case 'e' : Pf("Purple");
```

```
}
```

ASCII  
value = 101

Case 65:

Case 'A':

Error,  
duplicate  
Cases  
not valid

x value

o/p

3

Yellow

27

Black yellow

ASCII

value = 67 ← C

white

101

Purple

31

Blue Black yellow

11

Orange white

67

white

115

Black yellow





## Topic : Control Statements - 3

Example :

int x, sum = 0; ~~x 4 7 6 5~~

scanf("%d", &x);

switch(x)

```
{
    case 2: sum++;          sum=1
    case 9: sum+=3;         sum=4
    case 4: sum-=2;         sum=2
    case -5: sum+=4;        sum=6
    default: sum--; break;  sum=5
    case 1: sum*=2;
}
```

printf("%d", sum);

i/p: x = 2

o/p: 5

Example

i/p: 2

int i = 1, j = 2, k = 3, x, result = 1;

scanf("%d", &x);

switch(x)

```
{
    case 1: i = j + k; i = 5
            switch(i)
            {
                case 3: result++;          result=3
                case 5: result+=2; break;
                default: result--;
            }
    case 3: j = k + i; j = 8
            switch(j)
            {
                case 2: result+=3;
                default: result--; result=2
            }
            break;
}
```

case 5: k = i + j; switch(k) { case 3: result++; break; } printf("%d", result);







## Topic : Control Statements - 3



### Iterative Control Statements

Iteration : Repetition, Loop

- When one (or) more stmts to be executed repeatedly, Iterative control statements can be used.

- 3 Types : do-while, while, for

Syntax:

Post-test  
(or)  
Exit control  
loop

do  
{  
    stmt(s);  
} while (Expression);

min times = 1

True

Ex:

```
int i = 1; // 1 2 3 4 5
do
{
    pf("%d ", i);
    i++;
} while (i < 5);
```

o/p: 1 2 3 4

Entry control /  
Pre-test Loop

while (Expression)  
{  
    statement(s);  
}

min = 0

```
int i = 1;
while (i < 5)
{
    pf("%d ", i);
    i++;
}
```

o/p: 1 2 3 4

Always Executed  
one time / First time

Decision Making  
Exp

for (Exp1; Exp2; Exp3)  
{  
    statement(s);  
}

True

False

A, B, C, D, B, C, D,  
B, C, D, ... B

1 2 3 4 5

```
int i;
for (i = 1; i < 5; i++)
    pf("%d ", i);
```

o/p: 1 2 3 4





## Topic : Control Statements - 3



### Examples

① `int i=1, sum=0;`

`i`  
~~1~~ ~~2~~ ~~3~~ ~~4~~ ~~5~~  
~~6~~ ~~7~~ ~~8~~ ~~9~~ ~~10~~ ~~11~~

`sum`  
~~0~~ ~~2~~ ~~6~~ ~~12~~

`do`

`{ i++;`

`sum = sum + (i++);`

`} while (i <= 9);`

`printf("%d", sum);`

`// 30`

`i=2`

$\Rightarrow \text{sum} = 0 + 2 = 2$

`3 <= 9 True`

`i=4`

$\text{sum} = 2 + 4$

`5 <= 9 True`

`i=6`

$\text{sum} = 6 + 6$

`7 <= 9 True`

`i=8`

$\text{sum} = 12 + 8 = 20$

`9 <= 9 True`

`i=10`

$\text{sum} = 20 + 10 = 30$

`11 <= 9 False`

`o/p:`





## Topic : Control Statements - 3



(2)

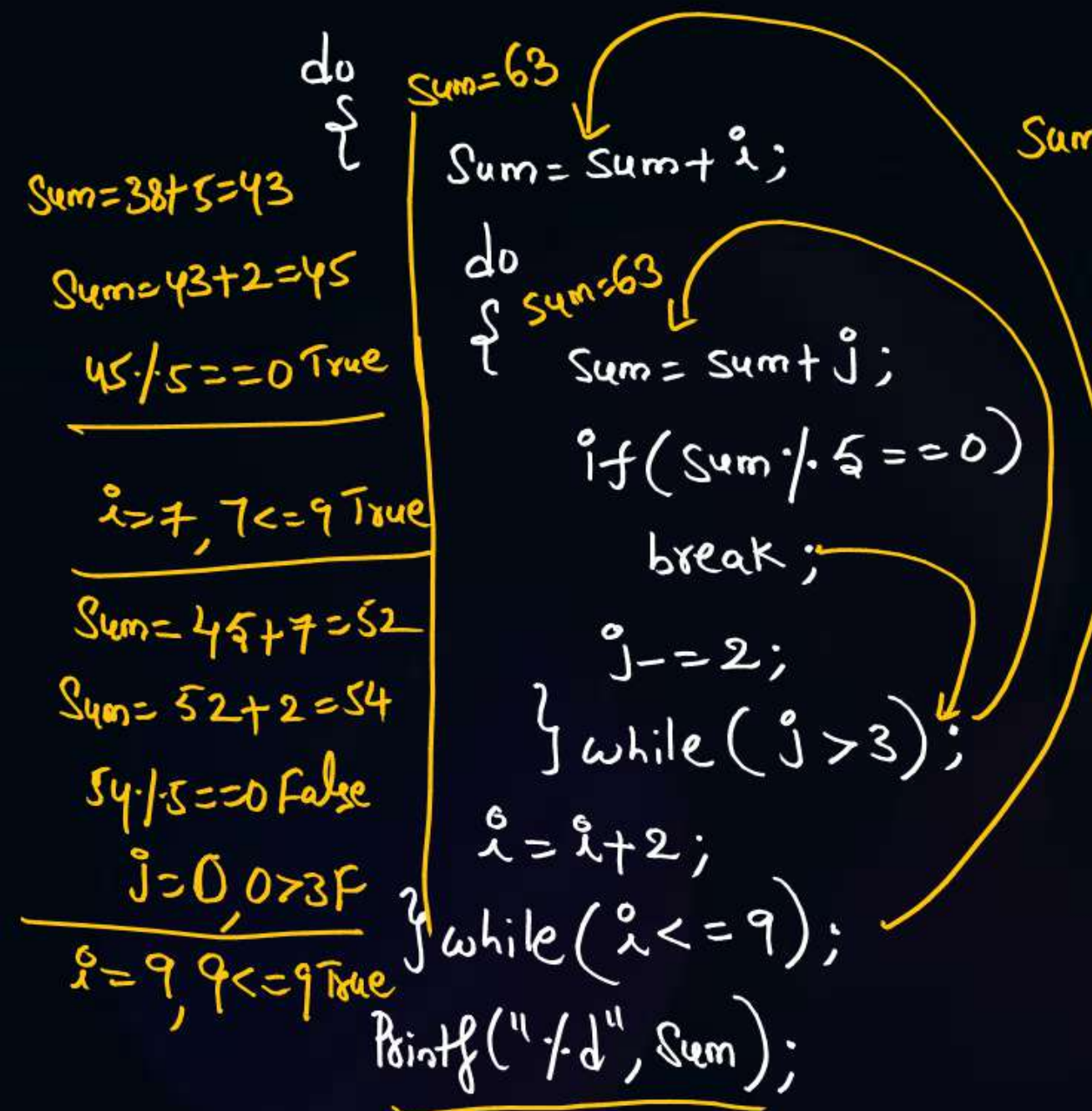
$$o/p = 63$$

int i=1, j=10, Sum=0;

i  
1 2 3 4 5 6 7 8 9  
11

j  
10 8 6 4 2  
-2

Sum  
0 11 19 25 34 38 43 45 52 54 63







## Topic : Control Statements - 3



### Home work

```
int main()  
{  
    int i=1, result=1;  
  
    do  
    {  
        result += i;  
        switch(i)  
        {  
            case 1: result = result + i;  
            case 3: result ++; break;  
            case 5: i = i + 1;  
            default: result = result - i;  
        }  
        i = i + 2;  
    } while(i <= 8);
```

```
printf("%d", result);  
return 0;  
}
```

o/p = \_\_\_\_\_





## 2 mins Summary



- Switch-Case
- Iterative Control stmts
  - do-while
  - while
  - for

To be Contd ...





**THANK - YOU**